

Edge Computing Smart Home Safety System

```
#define PIR_PIN 27

#define GREEN_LED 18

#define RED_LED 19

#define BUZZER 23

#define GAS_SENSOR 34


volatile bool motionDetected = false;


void IRAM_ATTR motionISR() {
    motionDetected = true;
}


void setup() {

    Serial.begin(115200);

    pinMode(PIR_PIN, INPUT);

    pinMode(GREEN_LED, OUTPUT);
    pinMode(RED_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);
```

```
attachInterrupt(  
  digitalPinToInterrupt(PIR_PIN),  
  motionISR,  
  RISING  
);
```

```
Serial.println("SMART HOME SAFETY SYSTEM");  
}
```

```
void loop() {
```

```
  int gasValue = analogRead(GAS_SENSOR);
```

```
  if(gasValue > 2500) {
```

```
    Serial.println("GAS LEAK DETECTED");
```

```
    digitalWrite(GREEN_LED, LOW);
```

```
    digitalWrite(RED_LED, HIGH);
```

```
    digitalWrite(BUZZER, HIGH);
```

```
    delay(500);
```

```
    digitalWrite(RED_LED, LOW);
```

```
    digitalWrite(BUZZER, LOW);
```

```
    delay(500);

    return;
}

if(motionDetected) {

    Serial.println("Motion Detected");

    digitalWrite(GREEN_LED, HIGH);

    delay(3000);

    digitalWrite(GREEN_LED, LOW);

    motionDetected = false;
}
}
```

Wokwi Link

<https://wokwi.com/projects/467254160597528577>